
Hop-depth spectra, scaling exponent, and entropy of literary syntax

Corpus construction and sampling. We selected 267 public-domain books in nine European languages from Project Gutenberg [1]: English (98), French (27), Spanish (26), German (26), Italian (25), Polish (19), Portuguese (19), Swedish (14), and Finnish (13). We downloaded each book as plain text (.txt). After we removed the Project Gutenberg licence banners, we discarded any file shorter than $C_{min} = 500$ characters. We kept a book only when spaCy (a grammatical parser that marks each word and the word it depends on) returned at least $N_{content} = 10,000$ content tokens (words that are not punctuation, spaces, or unreadable symbols). We curated English titles across novels, drama, verse, and non-fiction, and we removed duplicates. We dropped two Spanish books whose mean hop-depth was 5 or greater, because those parses had failed.

Prior text-to-series encodings. Earlier work already turns a text into a number series and takes its Fourier spectrum (the breakdown of a signal into frequencies): letter sequences [2], word-frequency rank [3], sentence-length variability (SLV; Drożdż et al. find literary $1/f^\beta$ with $\beta \approx 0.5$) [4], part-of-speech rank [5], and lexical information density [6]. Recursive grammars produce power-law mutual information; Markov-like strings do not [7]. We apply that construction to hop-depth [8]. We recompute SLV as a control: it is much flatter than hop-depth and is not interchangeable with it. Letter and rank series survive vocabulary-preserving shuffles; hop-depth does not. We compute spectral entropy as in Verma et al. [9], using Eqs. (1)–(2) of the main text. We report normalized composition Shannon entropy [10,11] separately, on the text itself rather than on the spectrum.

Syntactic hop-depth encoding. Hop-depth is the number of grammatical steps from a word to the main verb of its sentence (the root). Mean hierarchical distance (MHD) is the average of those steps [8]. MHD usually lies between about 2 and 4. We treat MHD as the mean of a token series and we analyse that series with a Fourier transform.

For each language we used spaCy with its largest Universal Dependencies model [12,13] (a shared map of which word depends on which). We used a medium or small model only when no large model existed. We turned off named-entity recognition (labelling names and places), lemmatization (reducing words to dictionary form), and text classification so the parser ran faster. We fed the text in blocks of $K = 50,000$ characters until we had collected $N_{content} = 10,000$ content tokens. We dropped tokens tagged PUNCT, SPACE, or X. For each remaining token t_i , hop-depth $x(i)$ is the number of head steps to the root. If a damaged parse looped and never reached a root, we stopped the walk at 100 steps so the count could not run away. MHD is the mean of that series at $N = 10,000$:

$$x(i) = \text{depth}(t_i), \quad \text{depth}(t) = \min[d : \text{head}^d(t) = t] \quad (\text{stop at } 100), \quad \text{MHD} = \frac{1}{N} \sum_{i=1}^N x(i).$$

A short example makes the encoding concrete. In *The cat sat on the mat.*, *sat* is the root. Walking each token to that root and dropping the period gives $x = [2, 1, 0, 1, 3, 2]$, so the sentence MHD is 1.50. Each book series is the concatenation of such sentences, truncated at N .

Hop-depth power spectral density. Each book is one sample per token, so frequency is $f = n/N$ in cycles per word ($n = 0, \dots, N - 1$), with Nyquist frequency $1/2$ and resolution $\Delta f = 10^{-4}$. We subtract the mean of the series and we take a single unwindowed discrete Fourier transform (DFT). We do not use Welch or short-time averaging, so the lowest frequency $f = 1/N$ remains resolved [4,3]:

$$\tilde{x} = x - \bar{x}, \quad X(f) = \sum_{n=0}^{N-1} \tilde{x}(n) e^{-2\pi i f n}, \quad E(f) = |X(f)|^2.$$

Subtracting the mean sends $E(0) \approx 0$. Because x is real, $E(f) = E(1 - f)$, and only the band $0 < f \leq 1/2$ is unique.

Spectral entropy and scaling. We obtain the scaling exponent β using Eq. (3) of the main text: ordinary least squares of $\log_{10} E(f)$ on $\log_{10} f$ after a 250-bin moving average over $0 < f \leq 1/2$. For the same spectrum we compute spectral entropy H using Eq. (1) with the mass function

$$P_f = \frac{E(f)}{\sum_f E(f)}.$$

Non-positive bins are dropped before the sum. We then report the unit-interval score $\tilde{H}$ from Eq. (2) with $N = 10,000$ ($\log_2 10,000 \approx 13.29$). Eq. (3) already implies $E(f) \propto f^{-\beta}$; $\beta = 1$ is pink noise. The spectrum is near-white for $f \lesssim 0.009 - 0.015$; linearly spaced bins still let the decaying band dominate Eq. (3). Restricting that fit to the decaying band steepens β by about 0.13–0.20 but does not change language rank. White, pink, and brown noise anchors give $\tilde{H} \approx 0.955, 0.747, \wedge 0.253$ via Eqs. (1)–(2). Literary hop-depth lies between white and pink.

Normalized composition Shannon entropy (real space). Separately we compute composition Shannon entropy on the book text itself (real space), not spectral entropy H from Eq. (1) and not an amplitude envelope. Each distinct Unicode character is a symbol, including spaces, punctuation, and case. We use the plug-in estimator [10,11]

$$S = - \sum_c p_c \log_2 p_c, \quad p_c = \frac{n_c}{N_{char}}.$$

This is the maximum-likelihood estimator of Shannon entropy [14]. Shannon used the same estimator for printed English [11]. At the character counts of a novel, plug-in bias is negligible at the precision we report [15]. Following Eq. (2) we report the unit-interval score

$$\tilde{S} = \frac{S}{\log_2 A}.$$

where A is the number of distinct characters. $\tilde{S}$ does not use the hop-depth series. We do not bin amplitudes and we do not sample an envelope $P(t)$.

The same cat-mat sentence used for MHD makes the count concrete. For *The cat sat on the mat.* we have $N_{char} = 23$ characters and 12 distinct symbols (Table 1). Then

$$S = -\frac{5}{23} \log_2 \frac{5}{23} - \frac{4}{23} \log_2 \frac{4}{23} - \dots = 3.29 \text{ bits/char}, \text{ so } \tilde{S} = 3.29 / \log_2 12 = 0.92.$$

A book-length English text lands near $\tilde{S} = 0.70$ because rare letters, digits, and punctuation enlarge A faster than they raise S . Language means of $\tilde{S}$ appear in Table 2.

Table 1. Real-space character counts for “The cat sat on the mat.” $S = 3.29$ bits/char; $\tilde{S} = 0.92$.

Symbol	Count n	p = n / 23	p (decimal)
space	5	5/23	0.217
t	4	4/23	0.174

a	3	3/23	0.130
e	2	2/23	0.087
h	2	2/23	0.087
T, c, m, n, o, s, .	1 each	1/23	0.043

Ablations. Unless we note otherwise, we used *Pride and Prejudice*, the English small parser, $E(f)$, and $N = 10,000$. White, pink, and brown noise anchors give $\tilde{H} \approx 0.955$, 0.747 , $\wedge 0.253$. Literary hop-depth lies between white and pink, and corpus MHD anti-correlates with $\tilde{H}$ (Pearson $r = -0.63$).

Sentence shuffle leaves the scaling almost unchanged ($s = -1.079 \rightarrow -1.066$; Shakespeare $\Delta s = -0.040$). Phase randomisation leaves $\tilde{H}$ unchanged. A part-of-speech-constrained word shuffle followed by re-parsing stays pink. SLV [4] is much flatter ($s \approx -0.20$) than hop-depth ($s \approx -1.08$). Subtracting each sentence’s mean depth yields $s = -0.938$, which is not white; per-sentence mean depth tracks SLV. On twelve texts, about $76\% \pm 5\%$ of $|s|$ is intra-sentence. Even forcing every sentence to length 23 remains pink ($s = -0.897$).

Cross-linguistic summary. Table 2 reports language means. β is Eq. (3), MHD is mean hop-depth, $\tilde{H}$ is Eq. (2), and $\tilde{S}$ is normalized composition Shannon.

Table 2. Language means ($N = 10,000$ hop-depths/book). β : Eq. (3). $\tilde{H}$: Eq. (2). $\tilde{S}$: normalized composition Shannon.

Language	Family	n	MHD	β	$\tilde{H}$	$\tilde{S}$
English	IE/Germanic	98	3.07	1.07	0.85	0.70
Spanish	IE/Romance	26	2.53	0.86	0.86	0.70
French	IE/Romance	27	2.44	0.85	0.86	0.69
Swedish	IE/Germanic	14	2.42	0.90	0.87	0.70
Finnish	Uralic	13	2.03	0.88	0.87	0.70
Portuguese	IE/Romance	19	2.38	0.81	0.88	0.68
German	IE/Germanic	26	2.40	0.91	0.88	0.72
Italian	IE/Romance	25	2.47	0.81	0.88	0.70
Polish	IE/Slavic	19	2.84	0.78	0.89	0.70

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